

Ambient Systems

Machine Learning Technology for Smart Buildings

The Problem

Building owners and operators often struggle to managing the indoor environment in real-time because of three underlying challenges.



Air Quality

Indoor air quality is essential for creating an optimal indoor environment, yet it is difficult to measure in real-time.

Technological Complexity

Building operations consist of complex legacy systems and machinery that controls cooling, ventilation, and other processes.

Competing Priorities

Different buildings have different system requirements based on the unique needs of occupants and operators.



Air Quality

Indoor air quality is calculated by measuring four parameters: temperature, relative humidity, CO_2, and contaminants.

All four parameters need to be within specified boundaries to create an optimal indoor environment based on unique building operations and occupant needs.

Challenge #1

Challenge #2

Technological Complexity

Building operators are increasingly faced with technology integration challenges and difficulties managing operations in-house.



Competing Priorities

Challenge #3



Healthcare

Minimize airborne infections

Offi ces

Increase worker comfort and productivity

Historical Buildings

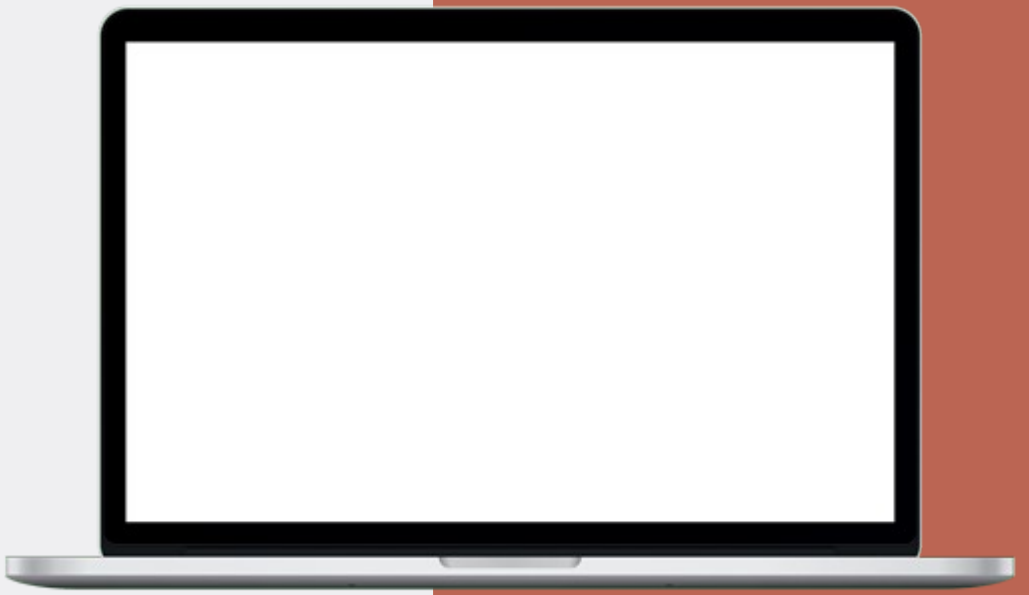
Preserve interiors of historical buildings and invaluable artifacts

Vertical Farms

Increase crop yield

Solution

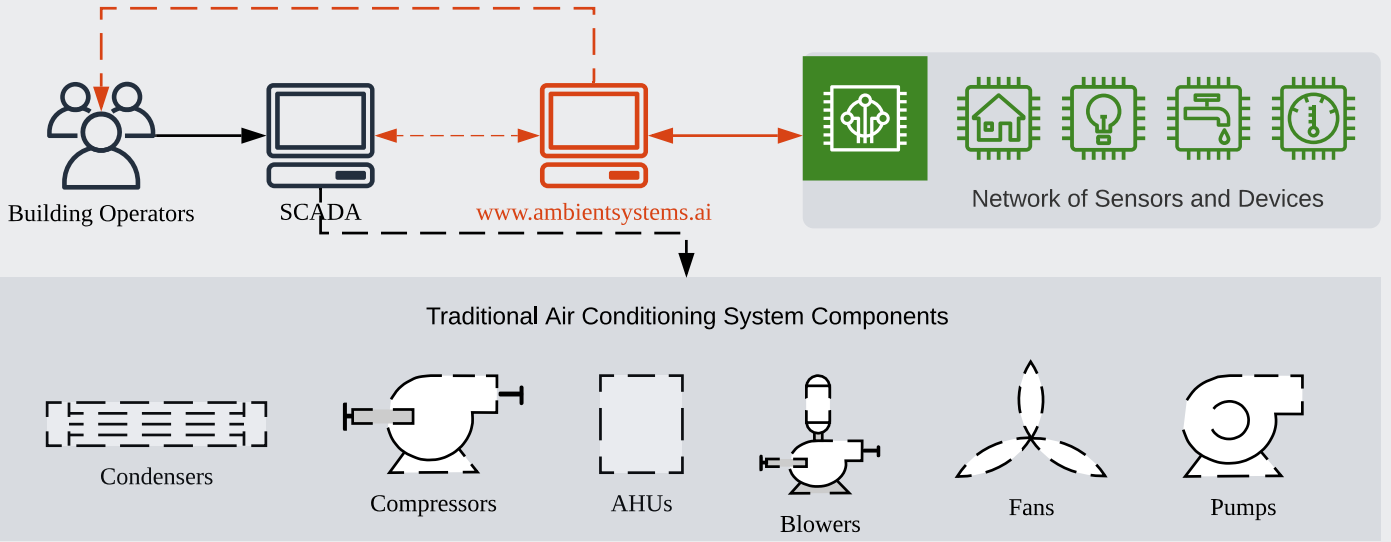
Ambient is a one stop shop for building owners and operators who want to manage indoor environment and air quality seamlessly and effectively.



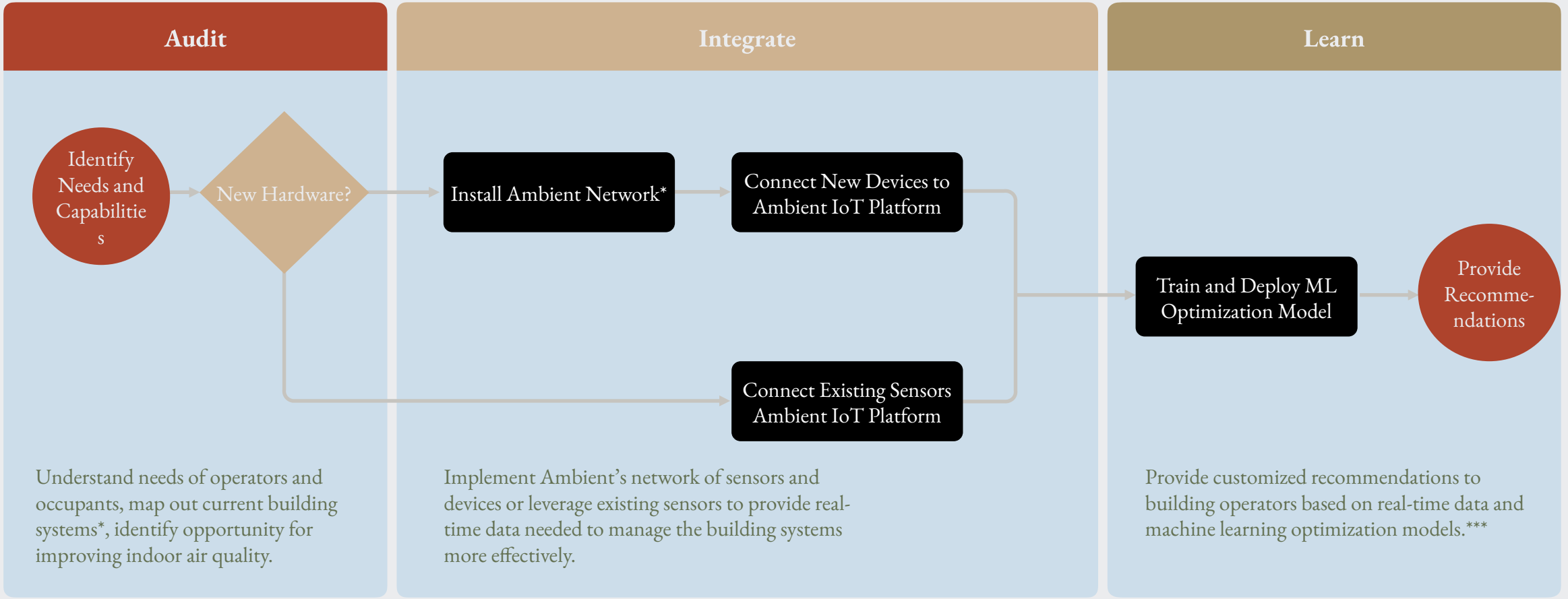
Ambient Systems High-level Schematic

What makes Ambient different?

- Technology tailored for unique building operations and occupant needs
- Seamlessly connects to the building's existing SCADA system and provides notifications directly to the operators
- Adds an additional layer of intelligence using AI-enabled optimization models
- Managed on the cloud and available across your devices
- Enabled by a network of interconnected sensors and devices (Internet-of-Things)
- Leverages NUS intellectual property to minimize maintenance and energy costs of sensors and devices



Process Diagram



* Focus is on building's systems related to managing indoor air quality, namely mechanical and electrical systems as well as relevant sensors and controls
** Based on unique building needs. Common input parameters: temperature, relative humidity, CO2, contaminants, occupancy, and user feedback.
*** Platform outputs are available online, delivered by email/message notifications, or connected to your building's SCADA system.

Benefits



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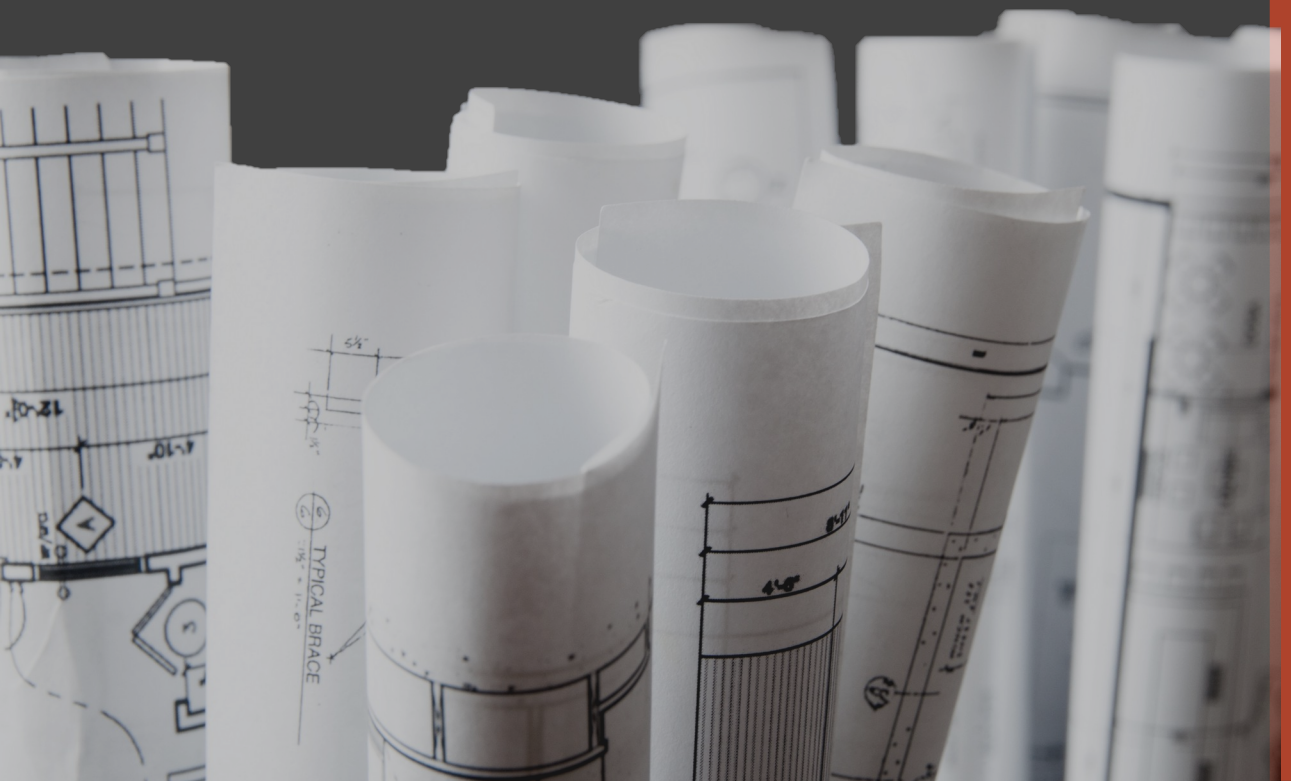
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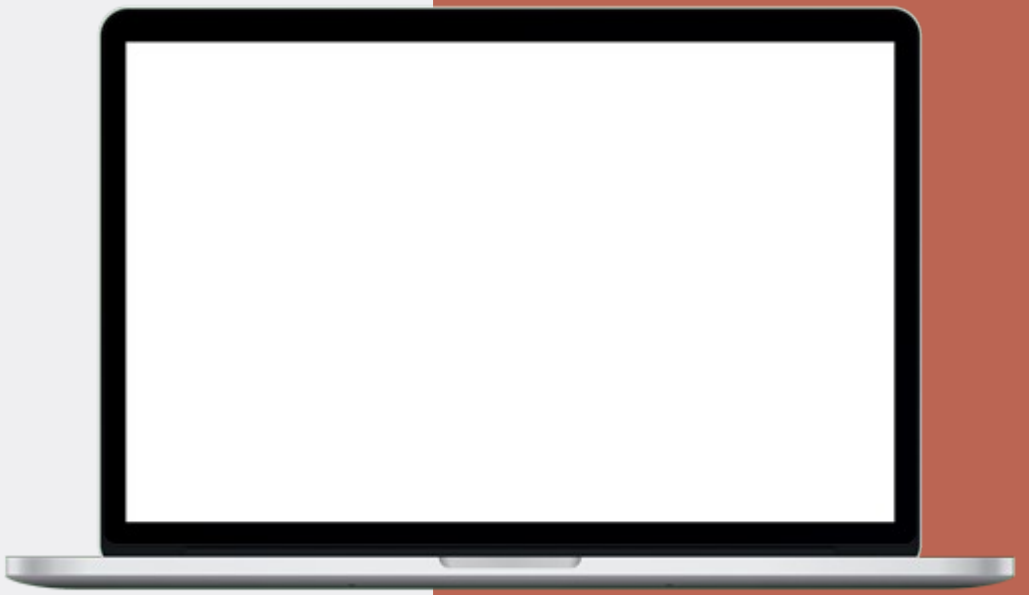
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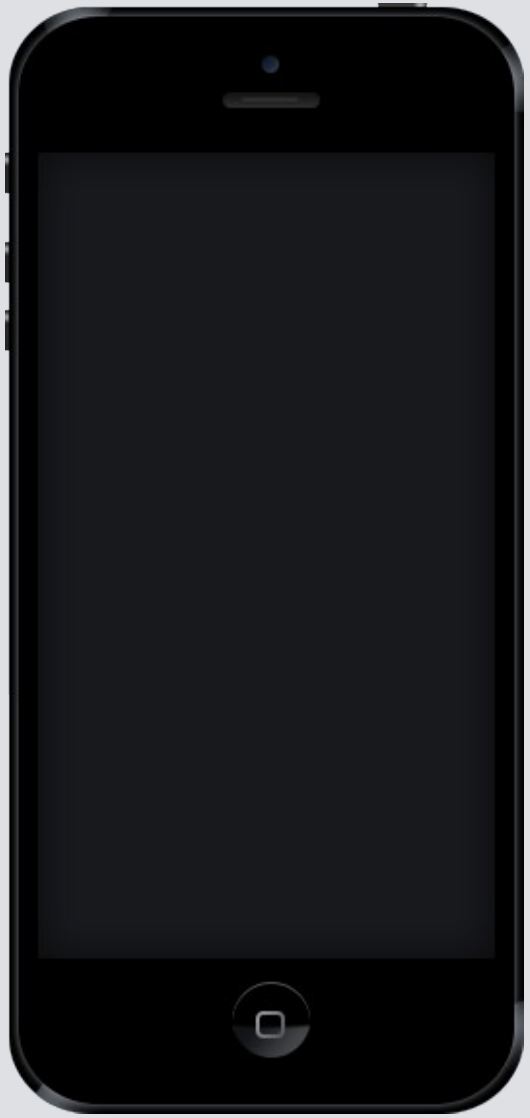
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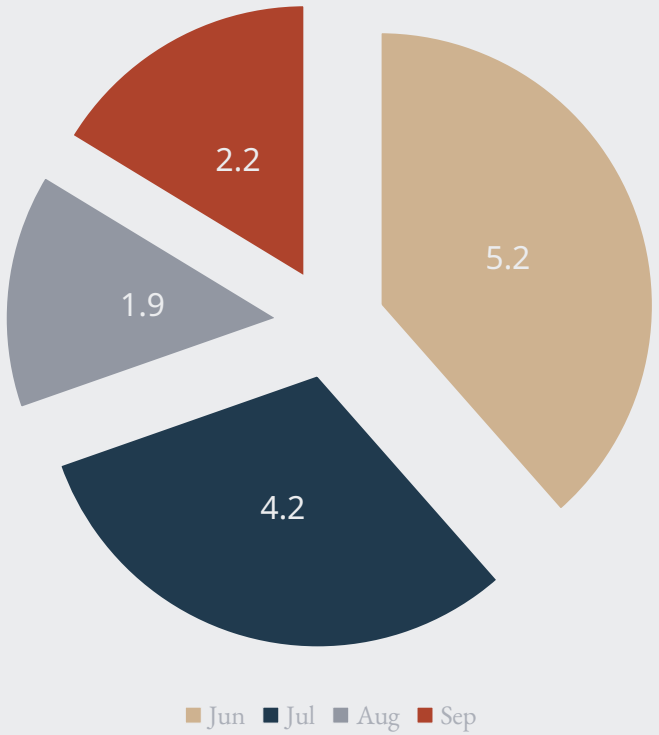
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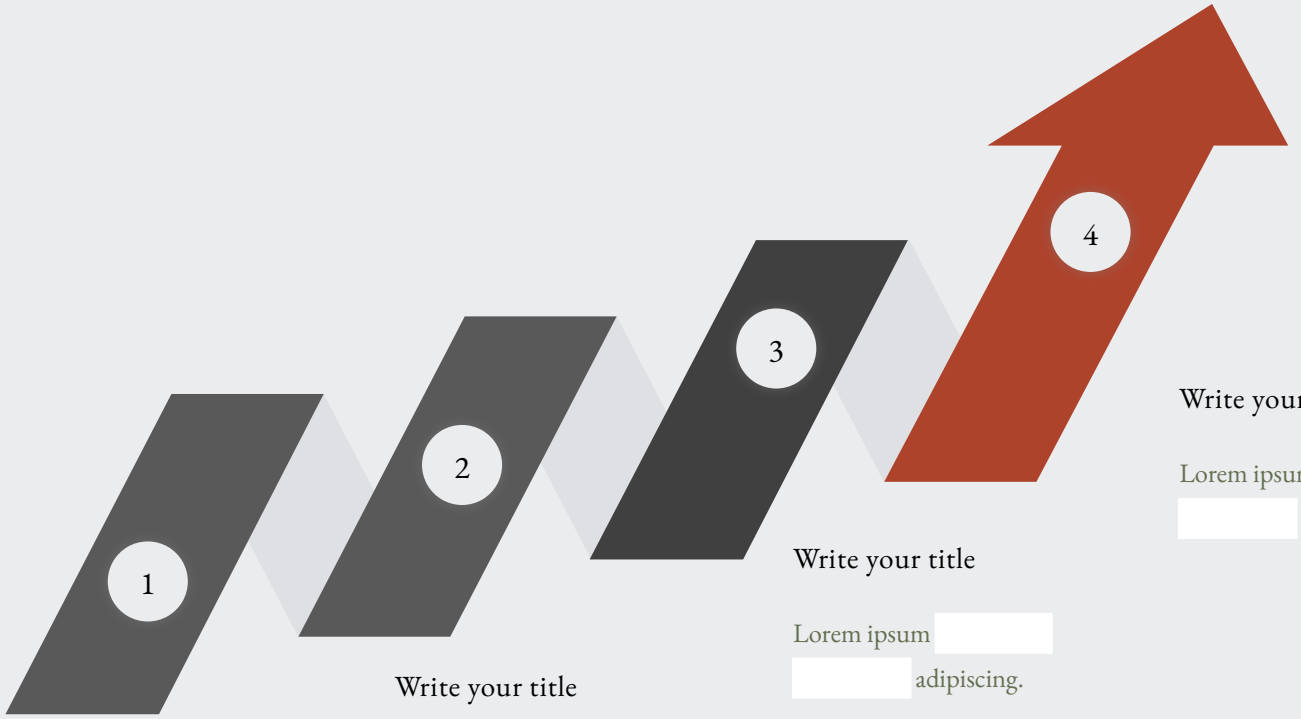
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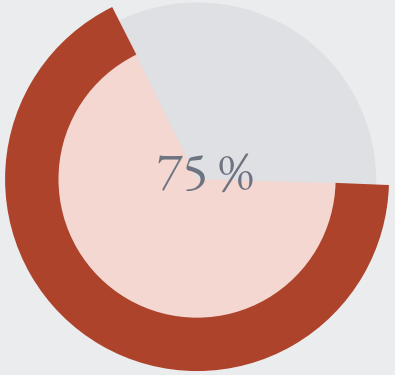
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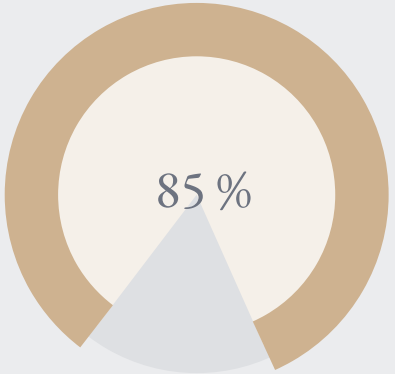
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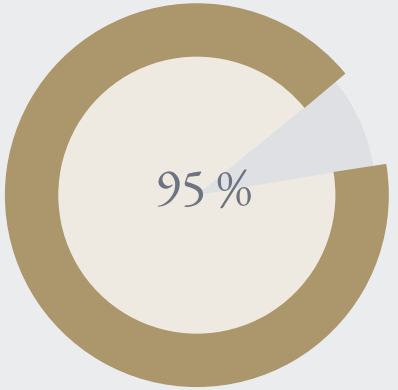
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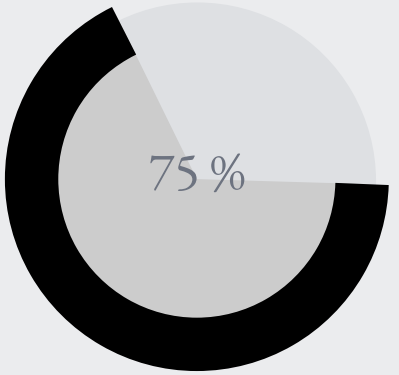
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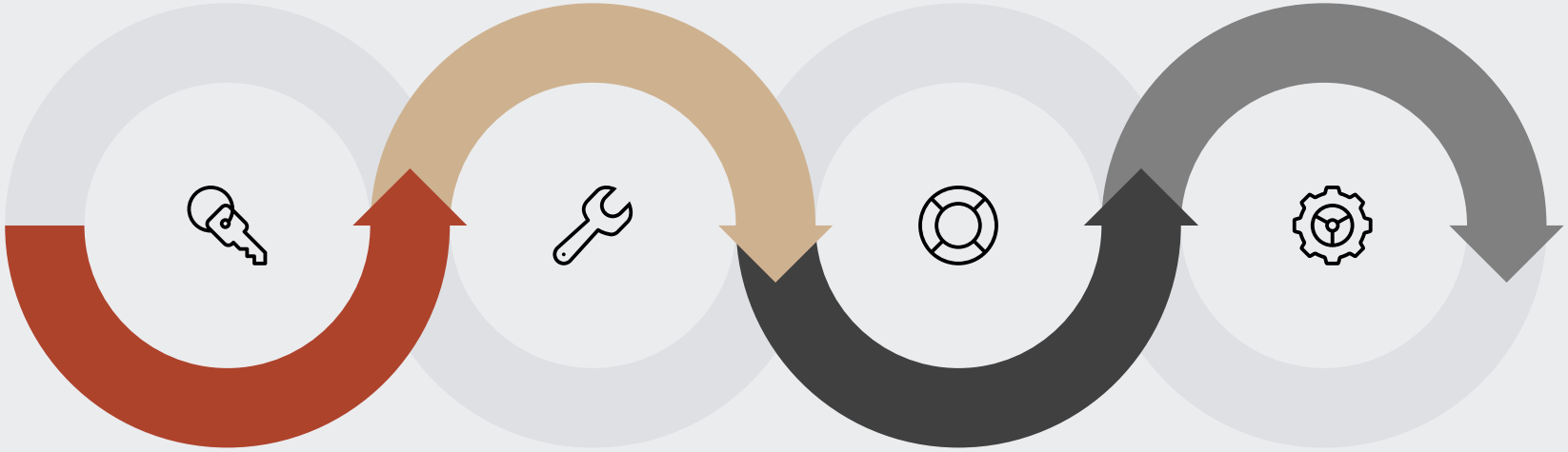
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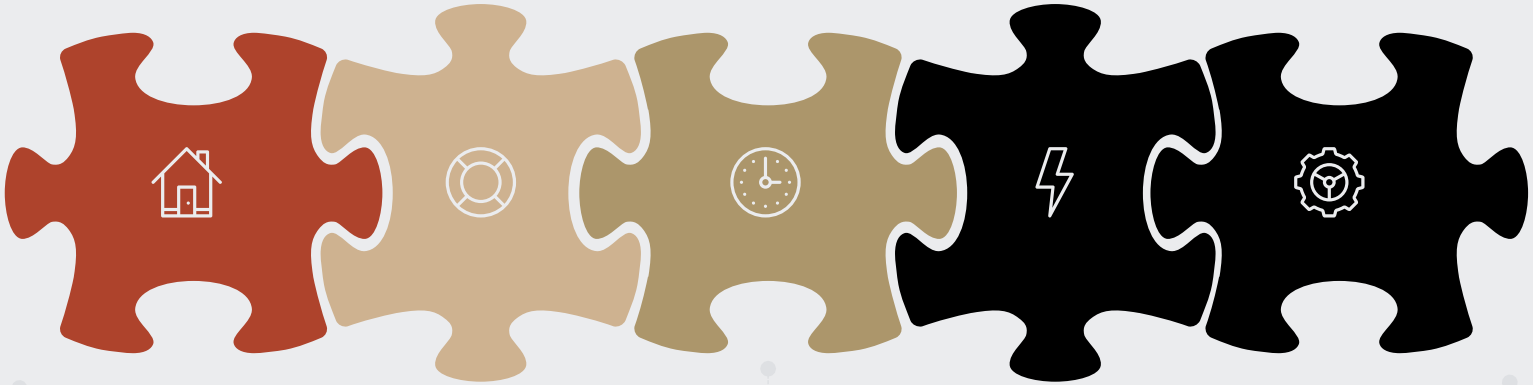
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